

ASSIGNMENT 2.2

Q1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

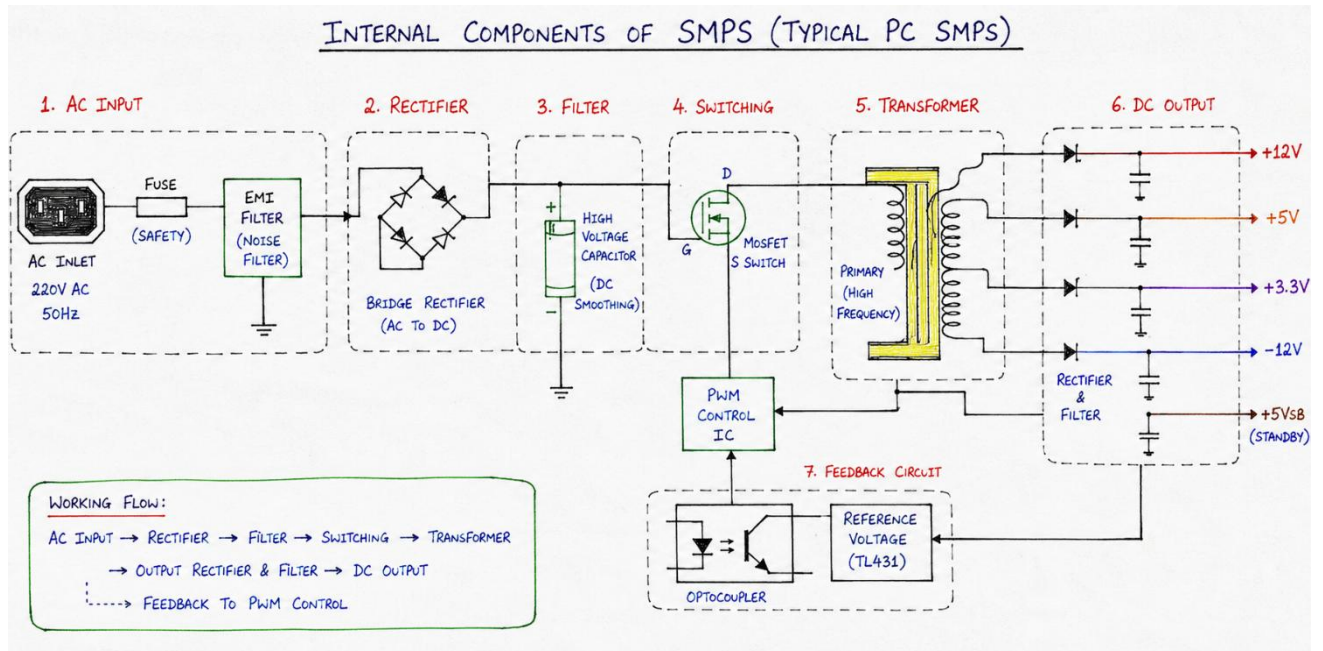
Devices Using SMPS (Switched Mode Power Supply)

- **Desktop PC:** It requires an SMPS to efficiently convert AC power into multiple stable DC voltages (3.3V, 5V, 12V) needed by different internal components like the CPU and GPU.
- **Smartphone Charger:** A tiny SMPS changes high wall voltage into safe 5V/9V DC power without melting the small plug.
- **LED Television:** A lightweight SMPS fits into flat screens to provide steady power and protect against voltage spikes.

Devices Using UPS (Uninterruptible Power Supply)

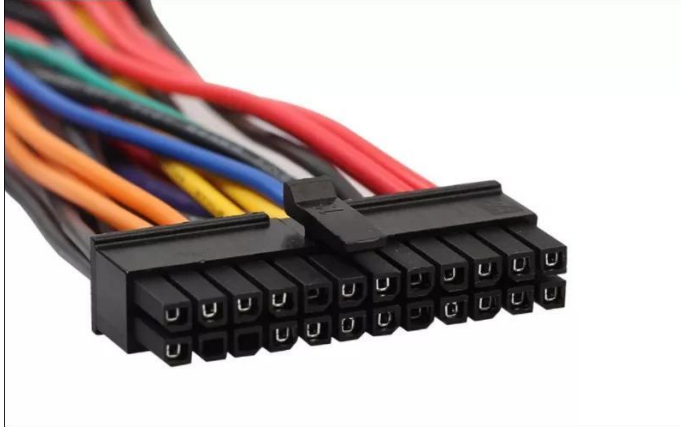
- **College Wi-Fi Router:** It connects to a UPS to ensure continuous internet connectivity and prevent network drops during sudden power outages.
- **Laboratory Desktop Computer:** It uses a UPS to provide temporary battery backup, giving users enough time to save critical research data and shut down safely during a blackout.

Q2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



Q3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

- **ATX 24-Pin:** Powers the **motherboard**.
Safety: Switch off and unplug the PC before connecting/disconnecting.



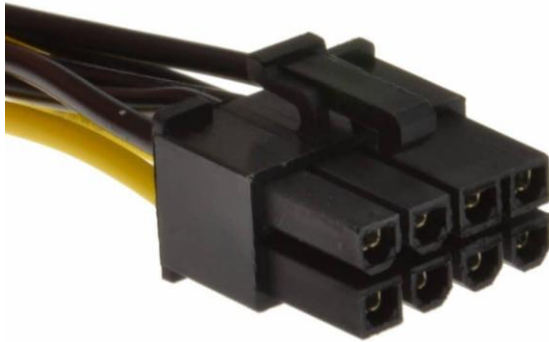
- **SATA Power:** Powers **HDD, SSD and optical drives**.
Safety: Match the L-shaped connector; never force it.



- **Molex:** Powers **older drives and case fans**.
Safety: Pull the connector by its plastic body, not by the wires.



- **PCIe 8-Pin:** Provides extra power to a **dedicated GPU**.
Safety: Ensure the connector is completely inserted and locked.



Q4. you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

➤ **Steps to Connect SMPS in a New Gaming PC:**

1. **Turn off the power** – Make sure the SMPS is switched OFF and the power cable is unplugged.
2. **Connect the motherboard** – Connect the **24-pin ATX cable** from the SMPS to the motherboard's **24-pin ATX power connector**.
3. **Connect CPU power** – Connect the **4+4-pin EPS/CPU cable** from the SMPS to the motherboard's **CPU power connector** near the CPU socket.
4. **Connect the graphics card** – Connect the **6-pin, 8-pin, or 6+2-pin PCIe cable** from the SMPS to the GPU's power connector.
5. **Connect the SATA storage device** – For a **SATA HDD or SATA SSD**, connect a **SATA power cable** from the SMPS to the drive.
6. **Check all connections** – Make sure every connector is fully inserted and its locking clip is properly secured.
7. **Final check and power on** – Ensure no loose cables are touching fans, then connect the AC power cable to the SMPS and switch it ON.